Joseph Walton

AI & Automation Lead | Lead Data Scientist | Machine Learning Engineer

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Professional Profile

AI and machine learning leader with over 10 years of experience designing, operationalising and improving data and AI products across central government, intellectual property, financial services and defence. Proven record of taking solutions from roadmap and evaluation through secure production deployment and adoption, delivering benefits of approximately £2M annually and major reductions in cloud cost and delivery time. Experienced in responsible AI, including evaluation, observability, human-in-the-loop workflows, red teaming and cross-government AI policy. Trusted communicator with technical, operational and international stakeholders, with experience leading delivery, mentoring colleagues and building organisational AI capability.

Core Expertise

- **AI strategy and delivery:** Product roadmaps, use-case assessment, solution architecture, generative AI, NLP, RAG, agents and recommendation systems
- **Operationalisation and automation:** MLOps, evaluation, observability, CI/CD, scalable data pipelines, multi-environment deployment and continuous improvement
- **Responsible AI and risk:** Red teaming, adversarial testing, human-in-the-loop evaluation, secure data handling and cross-government AI policy
- **Leadership and engagement:** Technical leadership, cross-functional delivery, senior stakeholder communication, international collaboration and mentoring
- **Technology:** Python, SQL, Azure AI Foundry, Azure OpenAI, LangChain, LangGraph, Azure Functions, Azure ML, GCP Vertex AI, MLflow, Elasticsearch, Polars, PySpark, TensorFlow and PyTorch

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Professional Experience

IBM

Lead Data Scientist
June 2025 - Present | Central government contract

- Rescued underperforming AI initiatives by introducing evidence-based evaluation and observability practices, improving the foundations for safe operational adoption.
- Designed and delivered an agent-based data processing solution using Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Developed a Claude Skill supporting human-in-the-loop ingestion for LLM evaluation data, saving approximately one hour per evaluation case.
- Re-engineered a data-matching algorithm from Apache Spark to Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10,000 per month.

Technology: Python, Polars, Azure AI Foundry, Azure Functions, LangGraph

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Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

- Set the product roadmap for AI applications in the intellectual property domain, balancing user value, technical feasibility, delivery constraints and risk.
- Designed and operationalised generative AI services using Azure OpenAI, LangChain and Azure Functions, including a RAG assistant that helped novice users identify relevant Goods and Services terms.
- Established an end-to-end MLOps approach across multiple environments, reducing model deployment time to under one day.
- Applied systematic evaluation to improve search and recommendation products, increasing relevance metrics by 20%.
- Conducted red-team exercises to assess generative AI applications against adversarial attacks and inform robustness improvements.
- Contributed to AI policy and best practice as a member of a cross-government AI working group.
- Built support for AI initiatives by explaining the IPO's approach to technical and non-technical stakeholders across multiple international IP offices.
- Architected Polars ETL pipelines to replace Azure Databricks workloads, reducing compute costs by 50% while enabling zero-downtime deployment.

Technology: Python, Azure OpenAI, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

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Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

- Led cross-functional delivery of production machine learning solutions using large-scale text and image data.
- Designed and implemented an MLOps framework on Google Cloud Platform, orchestrating training, evaluation, approval and deployment with Vertex AI Pipelines and reducing time to market to under one day.
- Developed and deployed NLP and computer vision products, including a multilingual BERT model that detected intellectual property infringements across more than 90 languages.
- Delivered a recommendation engine combining text and image similarity to improve infringement detection.
- Mentored a software engineer through a successful transition into a machine learning role, strengthening team capability.

Technology: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Designed and deployed a production NLP system in a secure, containerised environment, increasing enforced infringements by 10% per analyst across a workforce of approximately 200 analysts.
- Built and deployed an image-similarity model that helped secure a blue-chip client contract representing 10% of annual recurring revenue.

Technology: Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

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Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Developed machine learning models and scalable pipelines to securely link and process government administrative data in support of national surveys.
- Contributed to a cross-functional team delivering a highly scalable backend for the UK Census 2021 data collection system.

Technology: Python, PySpark, Cloudera

Barclays

Data Scientist (Secondment)

September 2018 - October 2018

- Applied statistical and analytical methods to securely and ethically use payments data for regional economic indicators, continuing work by the winning ONS x Barclays Data Hackathon team.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

- Developed full-stack, big-data processing systems for government and defence clients.

Technology: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Award

Winner, ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates of Gross Value Added.